

Shruti Murarka

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EDUCATION

Georgia Institute of Technology | **M.S. in Computational Science & Engineering** | **Atlanta, GA** Aug 2024 - May 2026
Coursework: Hardware-Software Codesign for ML, Data & Visual Analysis, HPC, DL, NLP (Graduate Teaching Assistant) **3.85/4 GPA**
National Institute of Technology (VNIT) | **B.Tech in Electronics & Communication Engineering** | **India** May 2018 - July 2022
Coursework: Machine Learning, OOP, OS, Image Analysis and Computer Vision, Linear Algebra & Probability **9.08/10 GPA** | **Rank: 4**

EXPERIENCE

Software Intern | **Generative Media, Amazon**, *Seattle, WA* Aug 2025 - Present

- Designing and developing a scalable system for **on-demand testing** of ML model pipelines, enabling Amazon Science teams to validate chains of models more efficiently.
- Implementing generalized input/output **AWS Lambda** functions to update **DynamoDB** tables and manage inputs in **S3** buckets.
- Building a **Harmony-based UI** that streamlines experimentation by allowing scientists to drag and drop components to generate dynamic **Airflow DAGs** and run ML models on **AWS Batch**.

AI Intern | **Biomedical AI, IBM Research**, *Yorktown Heights, NY* May 2025 - Aug 2025

- Built a modular **generation-augmented retrieval pipeline** integrating VAE-based molecule generation with synthesizability filtering, reducing infeasible molecules and saving \$10M in wet-lab screening costs.
- Engineered a **distributed inference system** for 1B+ molecules (PubChem, Savi, ZINC) using **Litdata Streaming Datasets, PyTorch Lightning, and DDP**, achieving scalable, high-throughput inference across multiple GPUs.
- Trained a **small-molecule foundation model** on 1B+ selfies data using transformers and VAEs; benchmarked on **MOSES**, providing performance insights for molecular generation (uniqueness:1, validity:1, novelty:0.999, sa-score:1.29).
- Deployed a **standalone Podman-based Milvus vector DB** indexing 80M+ PubChem molecules, enabling low-latency, synthesizability aware similarity search. Designed a **MCP server** and an integrated **Streamlit UI** for the GAR pipeline, supporting Agentic AI workflows.

Research Assistant | **Bio-MIBLab (Prof. May Wang)**, *Atlanta, GA* Sep 2024 - Jan 2025

- Developed an automated **long clinical notes summarization** system using large language models (LLMs) such as **BART, T5, GPT-4, BERTSUM**, and **Longformer** to enhance clinical workflow efficiency.
- Evaluated LLM's **faithfulness & factualness** with metrics like **QuestEval, Concept-F1, & SummaC**, for **Trustworthy AI**.

Software Development Engineer | **Natwest Group**, *Bangalore, India* July 2022 - July 2024

- Developed an **orchestrator service** leveraging multi-threading techniques, accelerating the customer's debit journey with API response time **less than 100ns**.
- Analyzed and optimized international transaction data, improving data retrieval efficiency by **30%** through advanced data structures and hashing techniques, enabling faster access and more efficient sorting/filtering of large datasets.
- Led **Agile** cross-functional teams to **migrate** a mainframe job monitoring tool to a React-based web app, streamlining approval/rejection processes across **4 banking platforms** and supporting **60K** daily users.
- Containerized microservices with **Docker** and orchestrated them using **AWS ECS**, reducing infrastructure costs by **20%** while improving fault tolerance across multi-node environments and application scalability.
- Mentored **5** junior software developers on SQL, Java and documented best practices within the organization.

RECENT HACKATHONS

ALPFA Recruitment — [Morgan Stanley's Code to Give Hackathon](#) Alpharetta, GA

- Developed an end-to-end job matching platform using **DistilBERT** word embeddings for relevance scoring and **Textkernel LLM API** for resume parsing. Deployed the model via **Flask API** with **Firestore** database, using **React & Vite.js** for the UI and **Amazon S3** for storage.

Bird Sound Recognition — [Kaggle's BirdCLEF 2024](#) Atlanta, GA

- Developed a deep learning model using **EfficientNet-B0** to identify **80+ bird species** from audio, improving performance by **30%** through noise reduction, and pseudo-labeling. Finetuned **Wav2Vec2** audio model achieving **0.7586** accuracy.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, NoSQL, React, Angular, HTML, CSS, Bash.

Machine Learning Stack: Pytorch, HuggingFace, Scikit Learn, Tensorflow, Keras, Numpy, Pandas, Matplotlib, OpenCV.

Libraries & Frameworks: MPI, Bootstrap, WebSDK, JUnit, SpringBoot (backend), Tyk Gateway, CI/CD, Flask.

Tools & others: PCF, AWS, MySQL, MongoDB, Chrome DevTools, VS Code, Postman, IntelliJ IDEA, Colab, Git, Agile.

PUBLICATIONS

- [[IEEE Access'23](#)]: A hierarchical approach for the diagnosis of sleep disorders using convolutional recurrent neural network.
- [[Computers in Biology and Medicine'22](#)]: Automated classification of cyclic alternating patterns sleep phases in healthy & sleep-disordered subjects using convolutional neural networks.

ACHIEVEMENTS

- JN Tata Endowment Scholar:** Among **Top 100** students in India to receive merit scholarship for study abroad.
- Hackathons:** Secured 2nd position - **Morgan Stanley's Code to Give Alpharetta Hackathon** | Secured 2nd position - **NatWest Challenge 23 Hackathon** | Awarded 'Business Acumen' & 'Innovation' honors - **NatWest Challenge 22 Hackathon**.
- Mentor** | **DeepLearning.AI:** Assisting over 5K learners in DL/ML specialization courses.